

Saudi Arabian Dates Market

Supply · Demand · Weather · Consumer Interest

A Data Science Analysis | 2000 – 2024

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ABSTRACT

This report presents a data-driven analysis of Saudi Arabia's date market, examining long-term production trends, weather influences, consumer search behaviour, and producer price dynamics across 2000–2024. Five statistical hypotheses were tested through OLS regression, Spearman and Pearson correlation, and a two-sample t-test, complemented by Random Forest and Ridge Regression machine learning models. Data was integrated from FAOSTAT agricultural records, the Open-Meteo historical weather API, and Google Trends search interest data. The analysis reveals that structural growth, trade dynamics, and consumer search patterns are stronger market determinants than weather variables alone, while Ramadan emerges as a statistically confirmed demand catalyst.

1. Research Overview

Saudi Arabia is one of the world's foremost producers of dates — a crop central to its agricultural economy and cultural identity. Despite their importance, the underlying drivers of production growth, price fluctuations, and demand patterns have rarely been examined through a rigorous, multi-source data lens. This project integrates four heterogeneous datasets to test five hypotheses spanning production trends, weather effects, religious demand cycles, and market pricing dynamics.

To what extent do time, weather, search interest, and production influence the supply and demand of dates in Saudi Arabia?

Four datasets were integrated: FAOSTAT agricultural records (production, yield, trade, and producer prices 2000–2024), the Open-Meteo historical weather API (daily temperature and precipitation aggregated seasonally), Google Trends search interest for 'dates' in Saudi Arabia (2004–2025), and public calendar data encoding Ramadan periods as binary demand indicators.

2. Hypothesis Testing — Summary

Five hypotheses were evaluated using OLS regression, correlation tests, and a two-sample t-test.

Hypothesis	Method	Key Statistic	Verdict
H1 – Production has increased over time	OLS Regression	$R^2=0.705$, $p<0.001$	✓ Confirmed
H2 – Weather influences production	OLS Regression	$R^2=0.782$, trend $p=0.030$	~ Partial
H3 – Ramadan increases search interest	Two-sample t-test	$t=2.24$, $p=0.026$	✓ Confirmed
H4 – Higher production lowers prices	Spearman Correlation	$\rho=-0.276$, $p=0.254$	✗ Not Confirmed
H5 – Search interest reflects demand	Pearson Correlation	$r=0.833$, $p<0.001$	✓ Confirmed

H1 — Long-Term Production Trend

An OLS regression of log-transformed production on year yielded $R^2 = 0.705$ with the year coefficient statistically significant ($p < 0.001$, $\text{coef} = 0.0327$). Production grew from approximately 735,000 tonnes in 2000 to over 1.9 million tonnes by 2024 — a 2.6× increase. The implied compound annual growth rate is approximately 3.3%. A notable outlier in 2014 (−40% YoY) is consistent with an abnormal harvest year; the broader structural trend remains stable throughout. Residuals were approximately normally distributed, suggesting an adequate model fit, though the small sample size ($n = 24$) limits the strength of this inference.

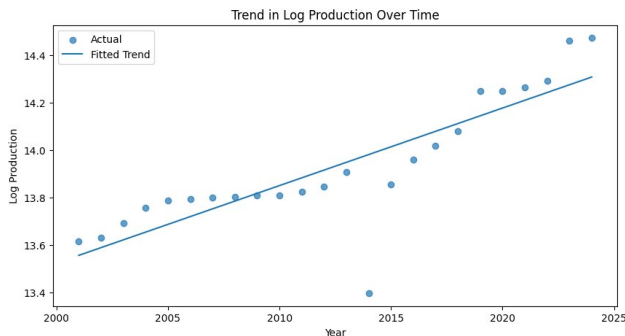


Fig. 1 — Log-transformed production (dots) with OLS fitted trend line ($R^2 = 0.705$, $p < 0.001$). The consistent upward trajectory confirms a statistically significant long-term growth trend, with the 2014 outlier clearly visible below the trend line.

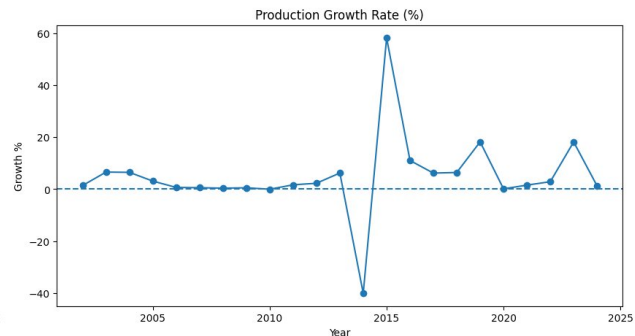


Fig. 2 — Annual production growth rate (%). Growth was stable at 2–7% through 2013, followed by a sharp −40% contraction in 2014 and a +58% rebound in 2015, before returning to moderate growth.

H2 — Weather Influence on Production

A multi-variable OLS model incorporating summer average temperature, fall precipitation, and a year trend variable achieved $R^2 = 0.782$. However, only the year trend reached the 5% significance threshold ($p = 0.030$); summer temperature was borderline ($p = 0.055$); fall precipitation showed no significant association ($p = 0.856$). Among all weather variables derived from the Open-Meteo historical API, summer temperature showed the strongest positive association with production ($r = 0.83$), followed by fall temperature ($r = 0.82$). Precipitation variables showed weaker and inconsistent patterns. These correlations are descriptive and do not imply that temperature causes higher output; confounding with the long-term time trend cannot be ruled out.

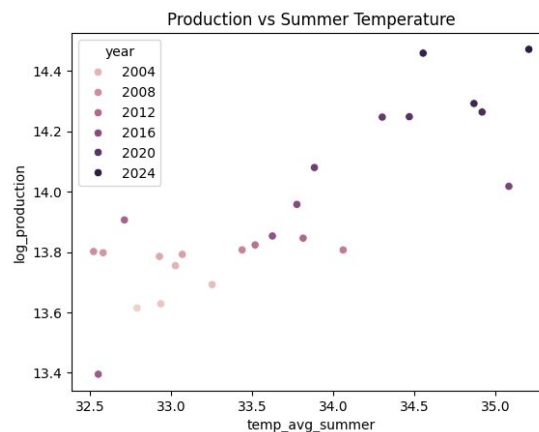


Fig. 3 — Scatter plot of log-transformed production against average summer temperature ($^{\circ}\text{C}$), with points coloured by year. Later years (darker purple) cluster toward higher temperatures and higher production values, reflecting the joint upward time trend in both variables rather than a direct causal temperature effect.

H3 — Ramadan Effect on Search Interest

Monthly Google Trends search interest for 'dates' in Saudi Arabia was analysed across 2015–2025, with eleven Ramadan periods encoded as binary indicators. An independent two-sample t-test returned $t = 2.24$, $p = 0.026$, indicating a statistically significant difference in mean search interest between Ramadan and non-Ramadan months. Mean search interest was approximately 42 during Ramadan compared to approximately 28 in non-Ramadan months. While this association is consistent with the cultural significance of dates during Ramadan, Google Trends data reflects relative search volume rather than actual consumption, and should be interpreted as an indirect demand signal.

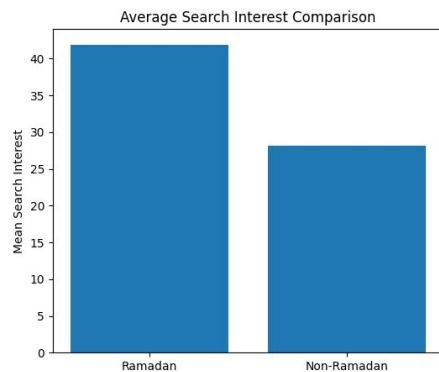


Fig. 4 — Mean search interest by period. Ramadan months average ~42 compared to ~28 for non-Ramadan months, a statistically significant difference ($t = 2.24$, $p = 0.026$).

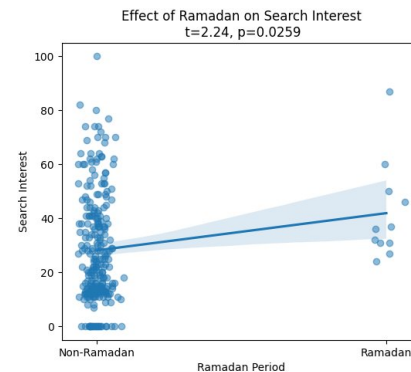


Fig. 5 — Scatter plot showing individual monthly search interest values for non-Ramadan and Ramadan periods, with regression line. The upward slope and t-statistic ($t = 2.24$, $p = 0.026$) confirm the significant elevation during Ramadan.

H4 — Production vs Producer Price

Spearman rank correlation between annual production and USD producer price yielded $\rho = -0.276$ ($p = 0.254$), which does not reach statistical significance. The evidence is therefore insufficient to conclude that higher production is associated with lower prices in this dataset. The dual-axis time series suggests a structural price break around 2014–2015, where prices fell from the 3,300–3,800 USD/tonne range to approximately 2,640 USD/tonne before partially recovering post-2020. This pattern is inconsistent with a simple supply-driven explanation and may be associated with broader macroeconomic conditions during that period, though no causal claim can be made from this data alone.

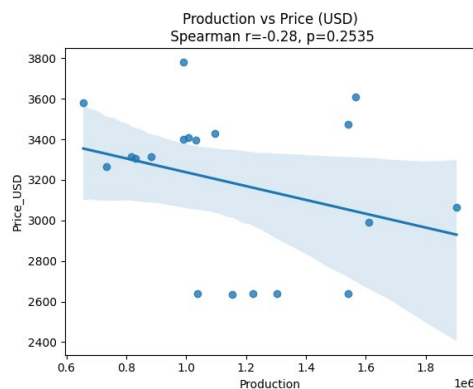


Fig. 6 — Scatter plot of production against producer price (USD/tonne) with Spearman regression line ($\rho = -0.28$, $p = 0.25$). The wide confidence interval and two visible price clusters suggest the relationship is not monotonic.

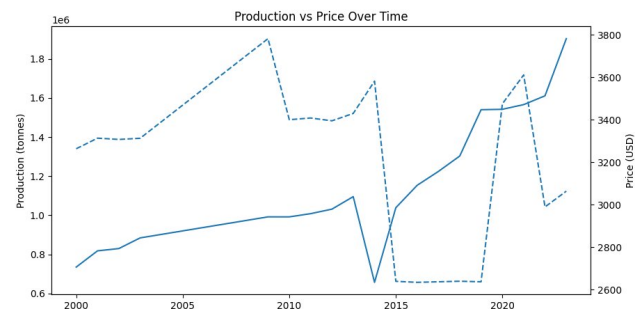


Fig. 7 — Dual-axis time series: production (solid line, left axis) and producer price (dashed line, right axis) 2000–2024. A pronounced price decline is visible around 2014–2015 despite continued production growth.

H5 — Search Interest as a Demand Proxy

Estimated annual demand — calculated as Production + Imports – Exports — was correlated against the yearly average of monthly Google Trends search interest. Pearson $r = 0.833$ ($p = 2.8 \times 10^{-6}$) indicates a strong positive linear association between the two series. Both variables show a broadly similar upward trajectory from 2004 onward, with a notable demand contraction in 2014 that is only partially reflected in search interest. While this association is strong, it is important to note that search interest is a proxy for consumer attention rather than a direct measure of consumption, and the estimated demand variable itself carries uncertainty from trade data quality.

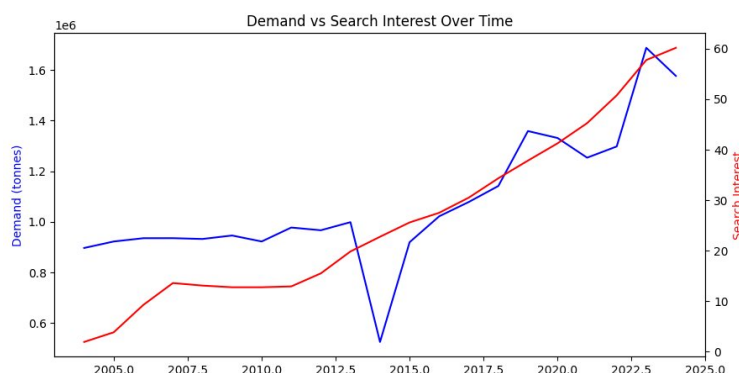


Fig. 8 — Estimated annual demand in tonnes (blue, left axis) and average Google Trends search interest (red, right axis) from 2004 to 2024. The two series show broad co-movement (Pearson $r = 0.833$, $p < 0.001$), particularly from 2015 onward, though the 2014 demand dip diverges from the continued rise in search interest.

3. Machine Learning — Producer Price Prediction

Random Forest and Ridge Regression were trained under Leave-One-Out Cross-Validation (LOOCV) across four thematic feature groups to predict log-transformed producer prices ($n \approx 19$ clean observations after merging all sources).

Group	Key Features	RF R^2	RF MAE	Ridge R^2
Supply	production_lag1, yield_lag1, year_trend	0.274	0.079	-0.059
Demand	export_qty_lag1, search_index, net_exports	0.305	0.084	-0.140
Weather	avg_temp_max, avg_temp_min, total_precip, precip_days	-0.182	0.099	0.026
Lag	producer_price_lag1, export_qty_lag1, search_index_lag1	0.030	0.102	-0.068

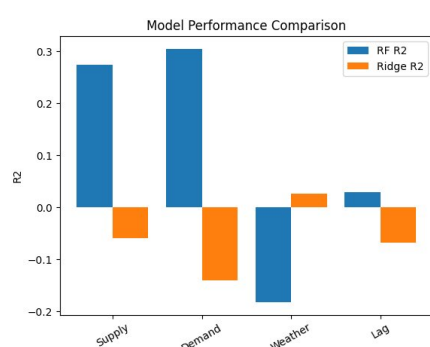


Fig. 9 — LOOCV R^2 scores for Random Forest (blue) and Ridge Regression (orange) across all four feature groups. RF outperforms Ridge in Supply and Demand groups; both models perform poorly on Weather and Lag features.

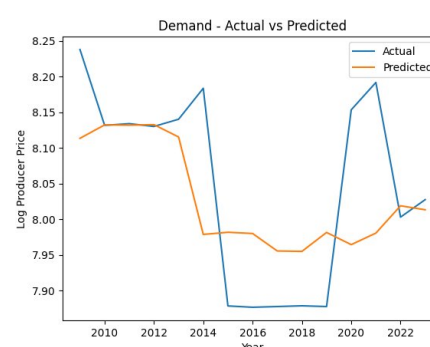


Fig. 10 — Actual vs predicted log producer price for the best-performing Demand model (RF). The model captures the general level but struggles with the 2014–2015 structural break.

4. Machine Learning — Feature Importance

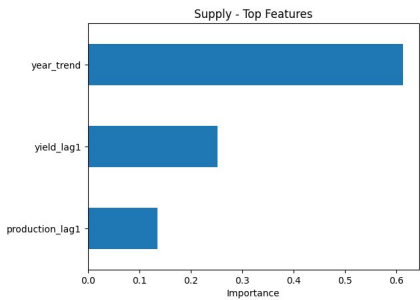


Fig. 11 — RF feature importance for the Supply group. *year_trend* accounts for 61% of importance, with *yield_lag1* (25%) and *production_lag1* (13%) contributing the remainder.

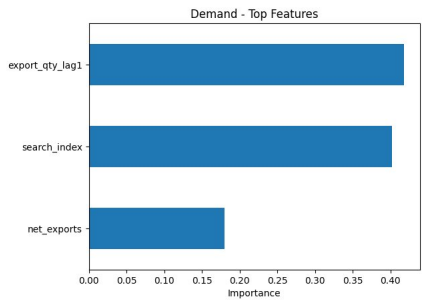


Fig. 12 — RF feature importance for the Demand group. *export_qty_lag1* (~42%) and *search_index* (~40%) are roughly equal in importance, with *net_exports* contributing ~18%.

Random Forest outperformed Ridge Regression across all feature groups under LOOCV, suggesting that the relationships between predictors and producer prices may be non-linear. The dominance of *year_trend* (61% importance in the Supply model) is consistent with the long-term structural trend identified in H1 and H2. In the Demand model, export quantities and search interest together account for approximately 83% of combined feature importance, indicating that trade-side and digital demand signals may be more informative for price prediction than weather or lagged supply variables. Weather features produced a negative RF R^2 (−0.182), performing below a naive mean predictor under LOOCV, which is consistent with the non-significant weather coefficients observed in the OLS regression for H2.

5. Key Findings

- 1

Production Growth Follows a Long-Term Structural Trend

Date production more than doubled between 2000 and 2024, with a statistically significant year trend ($p < 0.001$). This is consistent with sustained agricultural investment and area expansion, though year-level data limits causal attribution.

→ **Year trend is the strongest statistical predictor of production across all models.**
- 2

Weather Variables Show Associations but Lack Independent Significance

Temperature variables (especially summer, $r = 0.83$) are positively associated with production, but no weather variable reached 5% significance when controlling for year trend. Confounding with the time trend cannot be ruled out.

→ **Weather may contribute to production variation but cannot be isolated as an independent predictor.**
- 3

Search Interest is Significantly Higher During Ramadan

A two-sample t-test ($t = 2.24$, $p = 0.026$) indicates significantly elevated search interest during Ramadan. This is consistent with the cultural role of dates during the holy month, though search volume is a proxy for attention rather than consumption.

→ **The Ramadan period is associated with a statistically significant increase in search interest.**
- 4

No Significant Association Between Production Volume and Producer Price

Spearman $\rho = -0.276$ ($p = 0.254$) does not support a significant monotonic relationship. A structural price break around 2014–2015 suggests market factors beyond production volume are involved, though the specific drivers cannot be determined from this data.

→ **Production volume alone does not explain producer price variation in this dataset.**
- 5

Search Interest Strongly Co-moves with Estimated Demand

Pearson $r = 0.833$ ($p < 0.001$) indicates a strong linear association between Google Trends search interest and the estimated demand proxy. Both carry measurement uncertainty, but the strength of association suggests Google Trends may serve as a useful complementary indicator.

→ **Search interest shows strong co-movement with estimated demand and may serve as a demand proxy.**

6. Limitations

- **Small sample:** Yearly aggregation limits ML statistical power ($n \approx 19\text{--}24$ observations).
- **Demand proxy:** Google Trends reflects search intent, not actual purchase volumes.
- **Weather scope:** Seasonal averages may mask extreme events critical for harvests.
- **Data gaps:** FAOSTAT price gaps (2004–2008) reduce the ML training window.
- **Multicollinearity:** Year trend and temperature are correlated, raising condition numbers.
- **Geographic scale:** National weather averages ignore regional variation across growing areas.
- **Correlation $\neq$ cause:** All results are associational; causal inference needs quasi-experimental designs.

7. Conclusion

This study examined the factors associated with supply, demand, and pricing in Saudi Arabia's date market through integrated multi-source analysis. The results are broadly consistent across statistical and machine learning methods, though the small yearly dataset ($n \approx 19\text{--}24$ observations) limits the strength of causal inference.

Production shows a statistically significant long-term upward trend, consistent with structural agricultural investment. Search interest for dates is significantly higher during Ramadan, and shows a strong positive association with estimated annual demand ($r = 0.833$). In contrast, no significant association was found between production volume and producer prices, suggesting that price dynamics may be influenced by trade patterns, market structure, or broader economic conditions that fall outside the scope of this dataset.

Machine learning models indicated that demand-side variables — particularly export quantities and search interest — were more informative predictors of producer prices than weather or lagged supply variables. Weather features derived from the Open-Meteo historical API performed below a naive mean predictor under LOOCV, consistent with the non-significant weather coefficients in the OLS models. Random Forest outperformed Ridge Regression across all feature groups, suggesting possible non-linearity in the predictor–price relationships.

Taken together, these findings suggest that combining heterogeneous data sources — FAOSTAT agricultural records, Open-Meteo historical API weather data, and Google Trends search interest — can provide a richer basis for market analysis than any single source alone. The analytical approach used here is potentially applicable to other agricultural commodity markets where official data is sparse or subject to significant publication lag.

FAOSTAT: Food and Agriculture Organization — fao.org/faostat

Open-Meteo: Historical Weather API — open-meteo.com

Google Trends: trends.google.com · Keyword: 'dates' (Saudi Arabia)

Libraries: pandas · numpy · statsmodels · scikit-learn · scipy · matplotlib · seaborn

Environment: Google Colab / Jupyter Notebook

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